

1. Aim

To implement a simple **client-server chat application using UDP socket programming in Python.**

Algorithm – Server

1. Start the program.
2. Create a UDP socket using `socket()`.
3. Bind the socket to IP address localhost and port 9999.
4. Wait for a message from the client using `recvfrom()`.
5. Display the received message.
6. If the message is exit, terminate the chat.
7. Read a reply from the server user.
8. Send the reply to the client using `sendto()`.
9. Repeat steps 4–8.
10. Close the socket.
11. Stop.

Algorithm – Client

1. Start the program.
2. Create a UDP socket.
3. Specify the server address as localhost and port 9999.
4. Read a message from the client user.
5. Send the message to the server using `sendto()`.
6. If the message is exit, terminate the chat.
7. Receive the server's reply using `recvfrom()`.
8. Display the reply.
9. Repeat steps 4–8.
10. Close the socket.
11. Stop.

Program

Server

```
import socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_address = ("localhost", 9999)
server_socket.bind(server_address)
print("UDP Chat Server is ready...")
while True:
    message, client_address = server_socket.recvfrom(1024)
    message = message.decode()
    print("Client:", message)
    if message.lower() == "exit":
        print("Chat ended.")
        break
    reply = input("Server: ")
    server_socket.sendto(reply.encode(), client_address)
    if reply.lower() == "exit":
        break
server_socket.close()
```

Client

```
import socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_address = ("localhost", 9999)
print("UDP Chat Client is ready...")
while True:
    message = input("Client: ")
    client_socket.sendto(message.encode(), server_address)
    if message.lower() == "exit":
        break
    reply, server_address = client_socket.recvfrom(1024)
    print("Server:", reply.decode())
    if reply.decode().lower() == "exit":
        break
client_socket.close()
```